Project Background and Requirements for Custom PCB Functionality

Overview

We are embarking on a project that involves designing a custom 24V electronic PCB (control board) to manage the functionality of a TV mount system. The key feature is the use of two linear actuators that will lift the TV approximately 30 inches and then return it to its original position. The control board will interface with several peripherals to ensure smooth and safe operation.

Main Features and Peripherals

1. **Linear Actuators:**
    - **Specifications:** 2 x 12V 5A Linear Actuators with 5V potentiometers.
    - **Function:** The potentiometers provide analog voltage (0V-5V) indicating the position of the actuators. Fully retracted, the potentiometer wiper rests at 0.250V.
    - **Requirement:** The firmware must read the potentiometer voltage to determine the actuator's position.
2. **ToF Sensors:**
    - **Models:** Adafruit VL53L4CX or similar.
    - **Function:** Measure the distance from the top of the actuator to the sensor to monitor the actuator's position.
    - **Requirement:** Continuously monitor and report the position of the actuators.
3. **Human Presence Sensor:**
    - **Model:** STM STHS34OF80.
    - **Function:** Detects the presence of a person in front of the display.
    - **Requirement:** Initiates the actuators to bring the TV back down to its default position when a person is detected.
4. **IR Beam Break Sensor:**
    - **Function:** Acts as a safety mechanism.

- **Requirement:** If the TV is moving down and the beam is broken, the actuators must stop immediately to prevent injury or damage.

5. **Capacitive Touch Pad:**
    - **Function:** Used to start the TV experience.
    - **Requirement:** When pressed, the control board must start the experience sequence.
6. **GPIO (General Purpose Input/Output):**
    - **Function:** Communicates with an external media player device (Brightsign HD225).
    - **Requirement:** Send and receive high/low states to trigger actions like playing a video or displaying an image.
7. **LED Controller PCB:**
    - **Function:** Controls a 12V LED light strip (white lights).
    - **Requirement:** Blink the LEDs to indicate the actuators are active and the TV is moving, providing a visual cue for safety.
8. **Current Sensing:**
    - **Function:** Monitor the current draw of the actuators.
    - **Requirement:** Detect abnormal current draw (e.g., if the TV is stuck or obstructed) and act accordingly.

Operational Information for Components

1. **Capacitive Touch Button (CTB):**
    - **Action:** When pressed, the firmware must initiate the experience sequence.
2. **IR Beam Break Sensor:**
    - **Action:** If the beam is broken while the TV is moving down, the firmware must stop the actuators immediately.
3. **Presence Detector:**
    - **Action:** Trigger the TV to move to the default position when a person is NOT detected.

4. **LED Controller PCB:**
    - **Action:** Flash the LEDs to indicate actuator activity and TV movement.
5. **STM ToF Sensor:**
    - **Action:** Provide distance measurements to monitor actuator travel.
6. **GPIO:**
    - **Action:** Trigger events on the media player based on high/low signals.
7. **Actuator Potentiometer:**
    - **Action:** Provide the position of the actuators via a 0V-5V analog signal.

Requirements for Firmware Developers

- Ensure the firmware accurately reads and processes signals from the potentiometers, ToF sensors, presence detector, and beam break sensor.
- Implement safety mechanisms to stop the actuators if the beam break sensor is tripped or if abnormal current draw is detected.
- Enable communication with the external media player through GPIO signals.
- Manage the LED controller to provide visual cues during TV movement.
- Coordinate the sequence of operations based on input from the capacitive touch pad and other sensors.
- Please see Firmware Operational Document (PDF)